Grade 6 Computer Science Solved Exercises - Balochistan Textbook Board

Balochistan Textbook Board (BTBB)

Unit 1: ICT Fundamentals

Exercise 1. Fill in the Blanks (page 24)

i. Integrated circuits were used for the first time in ____ generation computers.

third

ii. An ____ device is used to show the information generated by a computer to the user.

output

iii. A mouse is an input device it is look like a ____.

mouse

iv. A scanner is an ____ that scans documents and books.

input device

v. A ____ is an input device that is used to record audio calls.

microphone

vi. A ____ is an input device that identifies the unique physical characteristics of a person.

biometric sensor

vii. A ____ is an output device which prints electronic data on the paper.

printer

viii. The printed paper is called ____ copy.

hard

ix. The standard mouse has two buttons called ____.

Left click and Right click

x. CRD stands for ____. (Flagged: the book prints 'CRD' and no expansion of 'CRD' appears anywhere in this unit, so no single word can be fixed with certainty. The closest book terms are CRT, Cathode Ray Tube, a monitor type, and CD, Compact Disc, a storage device.)

Flagged, not filled: as printed, this item has no certain answer in the book. Do not guess one word for 'CRD'.

Exercise 2. Tick the correct choice (page 25)

i. ICT stands for ____.

- Information Computer Technology
- Internal Computer Technology
- ✓ Information Communication Technology
- Integrated Communication Technology

ii. The first generation computers used ____.

- ✓ Vacuum Tubes
- Transistors
- IC
- Micro Chip

iii. The ____ secondary storage device is faster than HDD.

- RAM
- ✓ SSD
- CD
- Floppy Disk

iv. USB stands for ____.

- Unique Source Bus
- ✓ Universal Serial Bus
- Universal Super Bus
- Unique Serial Bus

v. The main circuit board inside the system unit is called ____.

- Processor
- RAM
- ROM
- ✓ Motherboard

vi. Processed data is called ____.

- Table
- ✓ information
- program
- hard copy

vii. Set of instructions is called ____.

- Data
- ✓ program

hardware

information

viii. Standard keyboard is also called ____.

ASDFG

POIUY

✓ QWERTY

ZXCVB

ix. Which of the following is not an input device?

keyboard

mouse

✓ printer

camera

x. ____ is an input device to scan physical parts of a person.

scanner

digital camera

✓ biometric sensor

microphone

Exercise 3. True or False - rewrite the statements to make them true (pages 25-26)

i. A mouse is an output device it is look like a mouse. False

ii. A printer is an input device that scans documents such as photographs. False

iii. LCD is a traditional monitor bigger in size. False

iv. A scanner is an output device which prints electronic data on the paper. False

v. The printed paper is called soft copy. False

Corrected statements: (i) A mouse is an INPUT device, and it looks like a mouse. (ii) A SCANNER is an input device that scans documents such as photographs; a printer is an output device. (iii) CRT is the traditional monitor, bigger in size; LCD is smaller and consumes less power. (iv) A scanner is an INPUT device; a PRINTER is the output device which prints electronic data on paper. (v) The printed paper is called HARD copy.

Exercise 4. Define the following terms (page 26)

i. ICT

ICT stands for Information and Communication Technology. It refers to a connected system of interactive tools and technologies that are used to send and receive information, with a focus on communication technology.

ii. Computer

A computer is an electronic machine that can accept data (input), manipulate the data according to specified rules (process), produce results (output) and store the results for future use.

iii. Data

Data is a collection of raw facts and figures. It is unorganized and

not useful by itself, and it can be words, numbers, images (pictures) and sounds.

iv. Information

Information is how you understand the facts: organized, useful data. It is the result of data processing, that is, transforming raw data into meaningful output.

v. Input devices

An input device allows a user to enter data and commands into the memory of a computer. Common examples are the keyboard, mouse, microphone, digital camera, scanner and biometric sensor.

vi. Output devices

An output device is used to show the information generated by a computer to the user. Common examples are the printer, monitor, speakers and multimedia projector.

Exercise 5. Differentiate the followings (page 26)

i. Input devices and Output devices

Input devices enter data and commands into the computer's memory, for example the keyboard, mouse and scanner. Output devices show the information generated by the computer to the user, for example the monitor, printer and speakers.

ii. Hardware and Software

Hardware means the physical parts of the computer system that can be seen and touched, and it cannot perform any task without

software. Software is a set of instructions (a program) that tells the hardware how to do a job, and it cannot be executed without hardware.

iii. RAM and ROM

RAM (Random Access Memory) is volatile temporary memory: its contents are lost when the computer is switched off, it allows both reading and writing, and it is fast. ROM (Read Only Memory) is non-volatile permanent memory: its contents are retained even when the device is powered off, the user can only read it, and it is slower and cheaper than RAM.

iv. Data and Information

Data is a collection of raw, unorganized facts and figures that is not useful by itself. Information is organized, useful data: it is how you understand those facts, and it depends on data.

v. Printer and scanner

A printer is an output device which prints electronic data on paper; the printed paper is called hard copy. A scanner is an input device that scans documents such as photographs, pages of text and books and converts them into digital format.

vi. Scanner and Biometric sensor

A scanner is an input device that scans documents such as photographs, pages of text and books. A biometric sensor is an input device that identifies the unique physical characteristics of a person, for example by face, palm, fingerprint or retina/iris scan.

Exercise 6. Write short answers (page 26)

i. Define the term ICT.

ICT stands for Information and Communication Technology. It is a connected system of interactive tools and technologies used to send and receive information, similar to Information Technology but focused on communication technology.

ii. List names of any three input devices.

Keyboard, mouse and scanner. (Other correct answers from the book: microphone, digital camera, biometric sensor.)

iii. What is numeric pad?

The numeric pad is the block of keys on the right side of the keyboard for quick number entry. It consists of the digits 0 to 9, plus /, *, +, -, ., Num Lock and Enter.

iv. What is the use of Function keys?

Function keys are the 12 keys marked F1 to F12 at the top of the keyboard. Every function key has a different meaning in different software, but F1 is used for HELP in almost all software.

v. Define the function of Right Click on the mouse.

Right click means pointing to an item on the screen and pressing and releasing the right mouse button. It displays a pop-up menu with various options.

vi. Define RAM.

RAM stands for Random Access Memory. It is volatile memory: data is retained in RAM as long as the computer is on, but it is lost when the computer is switched off. It lets the computer read data quickly to run applications.

vii. How ROM is different than RAM.

ROM is non-volatile and permanent: its contents are retained even when the device is powered off, and the user can only read the information, not alter it. RAM is volatile and temporary: its contents are lost when the computer is switched off, both reading and writing are possible, and it is faster than ROM.

viii. What is the difference between CRT and LCD?

CRT (Cathode Ray Tube) is the traditional monitor: bigger in size and usually not available in the market now. LCD (Liquid Crystal Display) is smaller and more expensive than CRT, and it consumes less power.

ix. Why we use MODEM?

A modem is a communications device that enables computers to communicate via telephone lines, so that computer users can communicate and exchange data and information with another computer.

x. How SSD is better than HDD?

An SSD (Solid State Drive) is a storage device similar to a hard disk drive, but it can access data faster than an HDD. It also maintains data in a permanent state even without power.

Exercise 7. Label the Keyboard segments (page 26)

Label the keyboard diagram using: 1. Numeric keypad, 2. Typewriter keys, 3. Function Keys, 4. Cursor control keys.

Top box (the row of F1 to F12 keys): 3. Function Keys. Bottom-left long box (the main letter block): 2. Typewriter keys. Bottom-middle box (the arrow keys): 4. Cursor control keys. Bottom-right box (the 0 to 9 block): 1. Numeric keypad.

Exercise 8. Label the following devices (page 27)

Label the six devices pictured.

a. Printer (an output device that prints electronic data on paper). b. Monitor (a CRT monitor, an output device). c. Multimedia projector (an output device that shows the computer's output on a big screen or wall). d. RAM modules (memory). e. Hard disk drive (a secondary storage device). f. Processor / CPU (Intel Core i7, the main part of the computer where instructions are processed).

Lab Activities (page 27)

Practical 1: Make a chart showing different types of ICT devices. Guidance: divide the chart into input devices (keyboard, mouse, scanner, microphone, camera, biometric sensor), output devices (monitor, printer, speakers, projector), storage devices (hard disk, SSD, USB drive, memory card, CD, DVD) and communication devices (modem). There is no single fixed chart in the book.

Practical 2: Observe your school computer lab and find what types of ICT devices exist in your lab. Guidance: list what you actually see in your own lab (for example monitors, keyboards, mice, printers, a

scanner) under input, output and storage headings. The answer depends on your lab.

Practical 3: Scan your picture by using the scanner in your lab. Guidance: place your picture face down on the scanner, run the scanning software, and save the scanned image. This is a hands-on task with no written answer.

Practical 4: Print your scanned picture. Guidance: open the scanned picture on the computer and print it with the lab printer; the printed paper is called hard copy.

Practical 5: Detach the keyboard and mouse and then again attach them to the system unit. Guidance: with the computer switched off, unplug the keyboard and mouse cables (or USB plugs), then plug them back into the system unit and check that both work. This is a hands-on task with no written answer.

Unit 2: Introduction to Operating System

Exercise 1. Fill in the Blanks (page 44)

i. The shortcut key of copy anything is ____.

Ctrl + C

ii. ____ shows different icons like document, download, music, videos, hard drives, CD, and USB etc.

This PC

iii. ____ contains all your apps, settings, and files.

Start menu

iv. ____ are small images that represent program, files and folders.

Desktop icons

v. Taskbar is the most important part of the ____.

Desktop

vi. The Windows Desktop is the main screen of the ____.

operating system

vii. ____ contains icons of battery, Wi-Fi, volume etc.

Notification Area

viii. ____ is a special folder where deleted items are stored.

Recycle Bin

ix. This PC is the new name of ____.

My Computer

x. The short cut key for creation of new folder is ____.

Ctrl + Shift + N

Exercise 2. Tick the correct choice (pages 44-45)

i. An operating system (OS) is a ____ that manages hardware of the computer.

- Information Computer Technology
- ✓ system software
- application software
- graphical software

ii. Windows is an operating system designed by ____.

- Sun Inc.
- Google
- Apple
- ✓ Microsoft

iii. The Windows ____ is the main screen of the operating system.

- Start button
- Start menu
- Icon
- ✓ Desktop

iv. The ____ is located in the lower-left corner of the screen.

- Desktop
- Universal Serial Bus
- ✓ start menu
- Recycle Bin

v. ____ contains icons of battery, Wi-Fi, volume, Clock etc. (Flagged: no printed option fits; the unit's own answer is Notification Area.)
Flagged, not ticked: none of the four printed options is correct. The answer given by the unit is the Notification Area, which is not listed among the options (Desktop, Universal Serial Bus, start menu, Recycle Bin).

vi. ____ is a special folder or directory where deleted items are temporarily stored.

- Desktop
- Universal Serial Bus
- start menu
- ✓ Recycle Bin

vii. A ____ is a collection of data or information stored in a computer.

- Data
- program
- ✓ file
- folder

viii. ____ is a container, that contains files or other folders or both.

Data

program

file

✓ folder

ix. The shortcut key to create a new folder is ____.

Ctrl + Alt + N

Ctrl + V

Ctrl + C

✓ Ctrl + Shift + N

x. The shortcut key of pasting anything is ____.

Ctrl + Alt + N

✓ Ctrl + V

Ctrl + C

Ctrl + Shift + N

Exercise 3. True or False - rewrite the statements to make them true (page 45)

i. Windows is an operating system designed by Apple. False

ii. The Windows Icon is the main screen of the operating system. False

iii. The start menu is located in the lower-right corner of the screen. False

iv. This PC are small images that represent program, files and

folders. False

v. Delete is a special folder or directory where deleted items are temporarily stored. False

Corrected statements: (i) Windows is an operating system designed by Microsoft. (ii) The Windows DESKTOP is the main screen of the operating system; an icon is a small image. (iii) The start menu is located in the LOWER-LEFT corner of the screen. (iv) DESKTOP ICONS are small images that represent programs, files and folders; This PC shows documents, downloads, music, videos, hard drives, CD and USB. (v) The RECYCLE BIN is a special folder where deleted items are temporarily stored.

Exercise 4. Define the following terms (page 45)

i. File

A file is a collection of data or information stored in a computer.

ii. Folder

A folder is a container that contains files or other folders (called subfolders) or both.

iii. Start Button

The Start button in Windows 10 is a small button that displays the Windows logo and is always displayed at the left end of the Taskbar. It is a table of contents to all applications (programs) and folders, and gives fast access to open any application.

iv. This PC

In Windows 10, the name of the 'My Computer' icon is replaced by 'This PC'. It shows different icons like document, download, music, videos, hard drives, CD, and USB.

v. Recycle Bin

The Recycle Bin is a special folder or directory where deleted items are temporarily stored. Deleted files are not permanently removed from the hard drive, and files in the Recycle Bin can be restored to their original location.

vi. Desktop

The Windows Desktop is the main screen of the operating system. On it you have access to tools like the Start Menu, Taskbar and other icons.

Exercise 5. Differentiate the followings (page 46)

i. Notification Area and Taskbar

The Taskbar is the most important part of the Desktop: by default it sits at the bottom of the screen and consists of the Start Menu, several application icons and the Notification Area. The Notification Area is only the right end of the Taskbar; it contains icons of battery, Wi-Fi, volume, Clock and Calendar, and the action center, and provides status and notifications.

ii. Start button and Start menu

The Start button is the small Windows-logo button at the left end of the Taskbar; clicking it opens the Start menu. The Start menu is the menu itself, located in the lower-left corner of the screen,

containing all your apps, settings and files. Open it by clicking the Windows icon, pressing the Windows key, or pressing Ctrl + Esc.

iii. Cut and Copy

Copy reproduces a file or folder without removing the original: right-click, Copy (Ctrl + C), then Paste (Ctrl + V) at the destination. Cut (with Paste) moves the file or folder: the original is removed from its first location once pasted at the new location.

iv. file and folder

A file is a collection of data or information stored in a computer. A folder is a container that holds files, other folders (subfolders), or both, so files can be organized and found easily.

v. Icon and desktop (printed as 'iv.' a second time in the book; the intended number is v.)

The Desktop is the main screen of the operating system, where you have access to the Start Menu, Taskbar and other icons. An icon is a small image on the Desktop that represents a program, file or folder and gives quick access to frequently used items.

vi. Ctrl + C and Ctrl + V

Ctrl + C is the shortcut key to copy a selected file, folder or text, leaving the original in place. Ctrl + V is the shortcut key to paste the copied item at the new location.

Exercise 6. Write short answers (page 46)

i. Define windows.

Windows is an operating system designed by Microsoft. It allows you to use a computer through a graphical interface, and it is the most popular operating system for home computers; the book names Windows 11 as the newest version.

ii. How to make a file?

Double-click 'This PC', click 'Documents' in the left pane, and double-click the folder 'Class-6'. Right-click in the empty area, select 'New', then select 'Text Document'. A document named 'New Text Document' is created; rename it, for example 'computer-ch2'.

iii. Write steps to delete a file.

Select the file or folder you do not need more. Right-click its name or icon and choose Delete from the pop-up menu, or use the Delete button in the Ribbon. A confirmation window appears; click 'Yes'. The file is deleted (it goes to the Recycle Bin, where it can be restored).

iv. How to create folder on the desktop?

Right-click on an empty part of the Desktop, select 'New', then click 'Folder' (the first option). A new folder icon appears with the name 'New Folder'; type a new name for it. In File Explorer you can also press Ctrl + Shift + N to create a new folder.

v. What is the purpose of start menu?

The Start menu contains all your apps, settings and files, and it gives fast access to open any application in the computer.

vi. What is dragging.

Dragging means moving the mouse while keeping the mouse button pushed down over a file or folder, and moving it to a new location. Once you have moved the object where you want to place it, you lift up the mouse button to 'drop' the object in the new location.

vii. Why we need folders?

Folders are containers that hold files or other folders (subfolders) or both. They let us organize our files so that they can be stored, found and managed easily instead of lying mixed together.

viii. Without an operating system, computer cannot do anything, why?

A computer cannot work without an operating system because the operating system is the system software that manages the computer's memory and processes, as well as all of its hardware and software, and it allows you to communicate with the computer. Without it, no program can run and no hardware can be used.

ix. What is the purpose of icon?

An icon is a small image that represents a program, file or folder. Its purpose is to give quick access to frequently used programs, files and folders on the Desktop.

Exercise 7. Label the Desktop diagram (page 46)

Label the diagram using: 1. Notification, 2. Start, 3. Task bar. (The Search box is already labelled in the book's picture.)

The diagram shows the Windows desktop with the Start menu open and the Search box already labelled. Left box (arrow to the bottom bar): 3. Task bar. Middle box (arrow to the Windows logo button at the left end of the Taskbar): 2. Start. Right box (arrow to the right end of the Taskbar, where the battery, Wi-Fi, volume and clock icons sit): 1. Notification (Notification Area).

Lab Activities (page 47)

Practical 1: Make a chart of the Windows Desktop. Guidance: draw the Desktop with its main parts labelled: Desktop icons (This PC, Recycle Bin), the Start button at the left end of the Taskbar, the Search box, the Taskbar at the bottom, and the Notification Area at the right end with battery, Wi-Fi, volume, clock and calendar.

Practical 2: Change the desktop background. Guidance: right-click on an empty part of the Desktop, choose Personalize (Display settings), and pick a new background picture. This is a hands-on task with no written answer.

Practical 3: Make a folder 'Pakistan' on the desktop. Guidance: right-click on the Desktop, select New, then Folder, and type the name 'Pakistan'.

Practical 4: Make a subfolder 'Punjab' in the 'Pakistan' folder. Guidance: double-click the 'Pakistan' folder to open it, then create a new folder inside it (right-click, New, Folder, or Ctrl + Shift + N) and name it 'Punjab'.

Practical 5: Copy your picture in the folder 'Panjab'. Guidance: select your picture, Copy it (Ctrl + C), open the 'Punjab' folder, and Paste (Ctrl + V). Note: the book spells the folder name 'Panjab' here although Practical 4 spells it 'Punjab'; use the same folder you

made in Practical 4.

Unit 3: Application Software

Exercise 1. Fill in the Blanks (page 67)

i. Microsoft Paint is a simple ____ editing program.

graphics

ii. The ____ shows the name of the software.

Title bar

iii. A ____ is a rectangular area that supports drawing and opens images in it.

Drawing area (Canvas)

iv. ____ is a set of available colors in a computer software.

Color Palette

v. ____ has many shapes, used for drawing different shapes.

Shapes tool

vi. ____ short cut key for saving an image.

Ctrl + S

vii. ____ short cut key is used to close a window.

Alt + F4

viii. Ctrl + C is used to ____.

copy

ix. Ctrl + V is used to ____.

paste

x. Most used font size is ____ points. (Flagged: the book states no font size for this item.)

Flagged, not filled: no font size is stated anywhere in this unit (the unit teaches Paint, Paint 3D and Prezi, and its summary lists Word shortcuts only), so no number of points can be fixed from the book.

Exercise 2. Tick the correct choice (pages 67-68)

i. MS Paint is a ____.

- Operating system
- system software
- ✓ application software
- right

ii. Quick Access Toolbar is located at the ____ of a window.

- Top
- bottom
- ✓ left
- Microsoft

iii. Rulers are used to ____ dimensions of the image.

- Color
- adjust
- ✓ measure

delete

iv. Foreground color is used for ____.

Body

text

shape

✓ border

v. ____ is used to add additional effects to an image.

Pen tool

eraser tool

✓ brush tool

color pallet

vi. ____ is used to save an image.

Ctrl + C

✓ Ctrl + S

Ctrl + V

Alt + S

vii. ____ is used to copy an image.

✓ Ctrl + C

Ctrl + S

Ctrl + V

Alt + S

viii. ____ is used to select the entire document.

✓ Ctrl + A

Ctrl + S

Ctrl + V

Alt + S

ix. The shortcut key to cut anything ____.

Ctrl + Alt + N

Ctrl + V

Ctrl + C

✓ Ctrl + X

x. To delete text to the right side press ____.

✓ Delete key

Ctrl + X

Backspace key

End key

Notes on MCQ ii and iv: (ii) The book says the Quick Access Toolbar is at the 'top left' of the Paint window; among the options, (c) left is the intended answer. (iv) The book's sentence says foreground colour is used for 'lines, borders of different shapes and text'; (d) border is ticked as the most distinctive option, but (b) text is also named in the same sentence.

Exercise 3. True or False - rewrite the statements to make them true (page 68)

i. To delete text to the right of the insertion point, press the Backspace key on your keyboard. False

ii. To delete text to the left of the insertion point, press the Delete

key on your keyboard. False

iii. Press Ctrl + X on your keyboard to copy anything. False

iv. Press Ctrl + C on your keyboard to cut anything. False

v. Press Ctrl + P on your keyboard to paste anything. False

Corrected statements: (i) To delete text to the RIGHT of the insertion point, press the DELETE key. (ii) To delete text to the LEFT of the insertion point, press the BACKSPACE key. (iii) Press Ctrl + C to copy anything; Ctrl + X cuts. (iv) Press Ctrl + X to cut anything; Ctrl + C copies. (v) Press Ctrl + V to paste anything; Ctrl + P is the print shortcut.

Exercise 4. Define the following terms (page 68)

i. Title bar

The title bar is a horizontal bar located at the top of a window. It displays the title of the software and the name of the current document.

ii. Ribbon

The Ribbon is a user interface element located below the Quick Access Toolbar and the Title Bar. Its purpose is to provide quick access to commonly used tasks. (The book also calls the tools pane a Toolbox: a window or pane containing icons and buttons that are tools in the program.)

iii. Color palette

The Color Palette is a set of available colors in a computer software. It appears in the Ribbon and provides two types of colors: foreground color and background color.

iv. Alignment (printed as 'vi.' in the book; the intended number is iv.)

Alignment means how text is lined up in a document, for example lined up at the left edge, centered, lined up at the right edge, or spread evenly (justified). (Flagged: the unit does not define Alignment; this is the standard Word meaning, given because the exercise asks for it.)

v. Bullate (Bullets)

Bullets (printed 'Bullate' in the book) are small symbols, such as dots, placed at the start of each item in a list to mark the items clearly. (Flagged: the unit does not define Bullets; this is the standard Word meaning, given because the exercise asks for it.)

vi. Ctrl + A

Ctrl + A is the shortcut key to select the entire document: click in the document and then press Ctrl + A.

Exercise 5. Differentiate the followings (page 68)

i. Pencil tool and Brush tool

The Pencil tool is used for free-hand drawing of an image: select Pencil in the Home tab, click and hold, and draw. The Brush tool is used to add additional effects to an image with various textures, such as calligraphy brush, air brush, oil brush, crayon, marker,

natural pencil and watercolor brush.

ii. Cut and Copy

Cut (Ctrl + X) removes the selected item from its place so it can be pasted elsewhere. Copy (Ctrl + C) reproduces the selected item without removing the original, so it can be pasted elsewhere while the original stays.

iii. Delete key and Backspace

The Delete key deletes text to the right of the insertion point. The Backspace key deletes text to the left of the insertion point (and, on the keyboard table in Unit 1, removes a character to the left of the cursor).

iv. Foreground color and Background color

Foreground color is used for lines, borders of different shapes and text. Background color is used to fill shapes and the background of the text or text boxes.

v. Save and Save As (printed as 'iv.' a second time in the book; the intended number is v.)

Save (Ctrl + S, or File, then Save) stores the file with its existing name and location. Save As opens the Save As dialog box so you can choose a new folder, a new file name, or a new format before saving.

Exercise 6. Write short answers (page 68)

i. Why MS Paint is good application for beginners?

MS Paint is a simple graphics editing program. It lets beginners create and edit pictures made up of dots on the screen with easy tools: pencil for free-hand drawing, brushes, shapes, fill with color, text and eraser, and it opens and saves common picture files (JPEG, GIF, PNG).

ii. What is the use of Pencil tool?

The Pencil tool is used for free-hand drawing of an image. In the Home tab select Pencil, then inside the drawing area click and hold the pencil tool and draw, for example a circle-like shape.

iii. What is the use of Brush tool?

The Brush tool is used to add additional effects to an image with various textures. Its types include calligraphy brush, air brush, oil brush, crayon, marker, natural pencil and watercolor brush.

iv. How to Write any text in a Circle?

First draw a circle with the Shapes tool: in the Home tab select the circle shape, then hold and drag the mouse to draw it. Then select the Text tool in the Home tab, click inside the circle so a text box appears, and type your text there.

v. Define MS Word.

Microsoft Word is a word processing application that allows you to create a variety of documents.

vi. How to save a document file?

Click File in the Menu bar and select Save, or press Ctrl + S. This

opens the Save As dialog box the first time: select the folder where you wish to save, enter the file name, and press the Save button.

vii. How to change color of a shape?

Select the shape, then in the Home tab use the Fill option to choose a fill colour and the Outline option to choose an outline colour from the list. Fill colours the inside of the shape; Outline colours its border.

viii. Write three names of shapes in Paint.

Rectangle, Ellipse and Triangle. (Other shapes named in the book: polygon, arrows, circle, lines.)

ix. How to select template in Prezi?

After logging in to Prezi, view your dashboard and select 'Prezi Presentation' on the left side of the screen. Click 'Create from template', scroll down, select a template according to your requirement, and click 'Use this Template'.

Exercise 7. Label the Paint window (page 69)

Label the diagram using: 1. Toolbox, 2. Handle, 3. Canvas, 4. Color Palette.

Top box (over the row of colours): 4. Color Palette. Middle box (over the large white area): 3. Canvas. Left box (over the column of tools): 1. Toolbox. Bottom-right box (at the corner of the canvas): 2. Handle (the resize handle used to change the canvas size).

Lab Activities (page 69)

Practical 1: Draw different types of triangles and color them. Guidance: in Paint, select a triangle shape from the Shapes tool, hold and drag to draw it, then fill it with a colour using the Fill option. Repeat for different triangle shapes. This is a hands-on drawing task with no single written answer.

Practical 2: Draw a circle and color it Orange color. Guidance: select the circle (ellipse) shape, hold and drag to draw it, then fill it with orange using the Fill option.

Practical 3: Draw an envelop. Guidance: draw a rectangle with the Shapes tool and add the flap lines as shown in the book's small picture, using lines from the Shapes tool.

Practical 4: Draw the shapes. Guidance: draw the shapes shown in the book's picture (for example a house scene built from a rectangle, triangle roof, circle sun and tree) using the Shapes tool, then colour them with Fill.

Practical 5: Draw a Dog. Guidance: use free-hand drawing with the Pencil tool, or build the dog from shapes (circles and ovals), as in the book's small picture, then colour it. This is a hands-on drawing task.

Unit 4: Internet

Exercise 1. Fill in the blanks (page 83)

(i) The ____ is global network.

Internet

(ii) The ARPANET stands for ____.

Advanced Research Projects Agency Network

(iii) NSF Stand for ____.

National Science Foundation

(iv) ____ is a collection of different websites on the internet.

The Web (World Wide Web)

(v) ____ is made up of related text, images and other resources.

A website

Exercise 2. Tick the correct choice (pages 83-84)

(i) You can directly go the web page through ____.

- Tab
- hyperlink
- ✓ URL
- gateway

(ii) Which term is used to save information from the internet?

- ✓ Downloading
- ISP
- Uploading
- Search Engine

(iii) Which term is used to have a connection for the internet?

- Downloading
- ✓ ISP
- Uploading
- Search Engine

(iv) Which term is used to send information to the internet?

- Downloading
- ISP
- ✓ Uploading
- Search Engine

(v) ____ make information easier to find on the internet.

- Downloading
- ISP
- Uploading
- ✓ Search Engine

(vi) ____ is the fifth generation of mobile network technology.

(Flagged: no printed option fits; the correct term is 5G.)

Flagged, not ticked: none of the four printed options is correct. The fifth generation of mobile network technology is 5G, which is not listed among the options (Tab, hyperlink, URL, gateway).

Exercise 3. True or False - rewrite the false statements to make them true (page 84)

(i) A server is a computer where you can search everything on the internet. False

(ii) Internet is a collection of different websites on the internet. False

(iii) Youtube make information easier to find. False

(iv) DSL service uses a broadband connection. True

(v) Downloading means sending data from your computer to remote computer. False

Corrected statements: (i) A SERVER is a computer where websites are stored; a SEARCH ENGINE is what you use to search everything on the internet. (ii) The WEB (World Wide Web) is a collection of different websites on the internet; the Internet itself is a global network of billions of computers and other electronic devices. (iii) SEARCH ENGINES make information easier to find; YouTube is a video website, not a search engine. (v) DOWNLOADING means receiving data from the internet onto your computer; UPLOADING means sending data from your computer to somewhere on the internet.

Exercise 4. Define the following terms (page 84)

(i) Internet

The Internet, also called the Net, is a global network of billions of computers and other electronic devices. It is the largest network of the world, a network of networks made up of millions of private, public, business and government networks.

(ii) URL

URL stands for Uniform Resource Locator. Each website has a unique address, called a URL; it is like a street address that tells your browser where to go on the Internet. Type it into the browser's address bar and press Enter to load the page.

(iii) www

WWW stands for World Wide Web, or Web. It is a collection of different websites on the Internet, developed in 1989 by Tim Berners-Lee.

(iv) Hyperlink (printed as '(iii)' a second time in the book; the intended number is iv.)

A hyperlink, or link, is any word or phrase on a website that is blue or underlined in blue. Links are used to navigate the Web: when you click a link, it usually takes you to a different webpage, and your cursor changes into a hand icon when you hover over a link.

(v) Uploading (printed as '(iv)' in the book; the intended number is v.)

Uploading means sending data or a file from your computer to somewhere on the Internet.

Exercise 5. Differentiate the following (page 84)

(i) Website and Internet

The Internet is the global network itself: a network of networks connecting billions of computers and devices. A website is one collection of related text, images and other resources that lives on the Internet; the Web is the collection of all such websites.

(ii) uploading and downloading

Uploading means sending data or a file from your computer to somewhere on the Internet. Downloading means receiving data or a file from the Internet onto your computer, for example clicking a link to a file that is then saved to your computer.

(iii) search engine and browser

A search engine (for example Google, Yahoo! or Bing) is a tool that finds information on the Internet: you type one or more keywords and it returns a list of relevant websites. A browser (for example Google Chrome, Safari, Mozilla Firefox or Internet Explorer) is the software that finds and displays websites; the browser itself is not the Internet, it only shows websites stored on the Internet.

(iv) Web site and web server

A website is made up of related text, images and other resources, and it has a purpose such as news, an online library or education. A web server is a computer where websites are stored: when you visit a website, your computer sends a request to the server, and the server retrieves the website and sends the correct data back.

Exercise 6. Write Short Answers (page 84)

(i) What is Internet?

The Internet, also called the Net, is a global network of billions of computers and other electronic devices. It is the largest network of the world, a network of networks of private, public, business and government networks, connected by physical cables (copper telephone wires, TV cables, fiber optic cables) and wireless connections like Wi-Fi and 3G/4G.

(ii) What is ISP?

ISP stands for Internet Service Provider. The type of Internet service you get depends on your ISP; common types are dial-up (slowest, uses a phone line), DSL (broadband, much faster than dial-up), cable (via cable TV), satellite (through satellites) and 3G/4G (wireless, mostly for mobile phones).

(iii) What is Web Browser?

A web browser is a type of software that allows you to find and view websites on the Internet. Common browsers are Google Chrome, Safari, Mozilla Firefox, Internet Explorer and Opera. Keep in mind that the browser itself is not the Internet; it only displays websites stored on the Internet.

(iv) Describe the function link?

A link (hyperlink) is a word or phrase on a website, usually blue or underlined in blue. Its function is navigation: when you click a link it usually takes you to a different webpage, and your cursor changes into a hand icon when you hover over it. Links can also point to a

file that can be downloaded.

(v) Describe the function of favorites?

Favorites, also known as bookmarks, save a website you want to view later so you do not have to memorize its exact web address. Locate and select the Star icon to bookmark the current website; your browser also keeps a history of every site you visit.

Q7. How can we search anything on the internet with the help of search engine?

Open a search engine such as Google, Yahoo! or Bing. Type one or more keywords (search terms) and press Enter. You will see a list of relevant websites, called search results; click a link to open a site that looks interesting, or return to the results page for more options. If the first try fails, try different keywords, use search suggestions, exclude a word with a minus sign (for example, cookies -chocolate), or put an exact phrase in quotation marks.

Q8. How can we download anything from the internet?

Find a link to the file you want: in some cases the link points to a file that can be downloaded, or saved, to your computer, and it may download automatically when you click it. To save an image, right-click the image and select 'Save image as' (or 'Save picture as').

Q9. Write a paragraph on 'how internet is useful for class 6 in the school'. (Model answer)

How the Internet is Useful for Class 6 in the School. (Model

paragraph) The Internet is very useful for Class 6 students in our school. With a web browser and a search engine, we can find information for our lessons, see pictures and videos that explain difficult topics, and learn new things beyond our textbook. We can download useful notes and images for projects, use email to send and receive messages, and check our work against authentic sources. But we must be careful: everything on the Internet is not from an authentic source, so we should consider the source, read to the end, analyze the facts, check other sources, and watch out for errors before we believe any information.

Lab Activities (page 85)

Practical 1: Use the search engine to view information about the Father of Computer, and bookmark the web pages that are the most interesting. Guidance: open a search engine, search for 'Father of Computer' (Charles Babbage), open the results, and bookmark the most interesting pages with the Star icon so you can find them again.

Practical 2: Download the interesting things about the Father of Computer, Charles Babbage, and write an essay of 100 words on the Father of Computer. Guidance: save the useful pages or pictures you found, then write your own 100-word essay about Charles Babbage from what you learned. The notes do not write your essay for you.

Practical 3: Search the recipe of cake excluding the word chocolate. Guidance: use the minus sign to exclude a word, for example search for: cake recipe -chocolate. This is the refining technique taught in section 4.6.3.

Practical 4: Search for impacts of using mobile phones in the age of under 10, and write advantages and disadvantages of mobile phones for school going students. Guidance: search with keywords such as 'mobile phones children under 10 advantages disadvantages', read more than one source, and write the advantages (for example learning and communication) and disadvantages (for example distraction and health concerns) in your own words.

Practical 5: Prepare a report on your school's computer lab, devices and peripherals found in lab. Guidance: visit your own school lab, list the devices you actually see (computers, monitors, keyboards, mice, printers, scanners, projectors) under input, output and storage headings, and write a short report. The answer depends on your own lab.

Unit 5: Algorithmic Thinking and Problem Solving

Exercise 1. Fill in the Blanks (page 95)

(i) ____ is a situation or an issue that needs to be solved.

Problem

(ii) ____ are straightforward and easy to understand.

Simple problems

(iii) ____ problems are more challenging.

Complex

(iv) ____ the problem is the process of figuring out what the problem is.

Identifying

(v) The word ____ is derived from the name of the mathematician, Mohammed ibn-Musa al-Khwarizmi.

algorithm

(vi) An algorithm is simply ____ to be followed.

a set of rules

(vii) Every step in an algorithm should be ____ and not confusing.

clear and precise

(viii) In algorithm writing the steps must be in the ____ to get the correct result.

right order

(ix) Writing an algorithm involves creating a ____ to solve a specific problem or perform a task.

step-by-step procedure or set of rules

(x) In writing of algorithm, the last step is ____.

End

Exercise 2. Tick the correct choice (pages 95-96)

(i) ____ is a situation or an issue that needs to be solved.

- Information
- Flowchart
- Algorithm
- ✓ Problem

(ii) Simple ____ are straightforward and easy to understand.

- simple
- complex
- finite
- ✓ Problems

(iii) ____ problems are more challenging and have many parts or steps.

- simple
- ✓ complex

finite

infinite

(iv) The process of writing a program is called ____.

✓ programming

testing

coding

debugging

(v) ____ statement is used to make decisions in a program.

programming

GOTO

coding

✓ IF

(vi) Step by step presentation of solution of a problem is called ____.

✓ algorithm

flowchart

program

hard copy

Exercise 3. True or False - rewrite the statements to make them true (page 96)

i. Program is a situation or an issue that needs to be solved. False

ii. complex problems are more challenging and have many parts or steps. True

iii. Programming is a skill to understand and solve any problem. False

iv. The process of writing a program is called Algorithm. False

v. The Last step of Algorithm is START. False

Corrected statements: (i) A PROBLEM is a situation or an issue that needs to be solved; a program is a set of instructions. (iii) Programming is a skill to understand and solve a COMPUTATIONAL problem (the book's own wording; 'any problem' is too wide). (iv) The process of writing a program is called PROGRAMMING (coding). (v) The last step of an algorithm is END; START is the first step.

Exercise 4. Define the following terms (page 96)

(i) Problem

A problem is a situation or an issue that needs to be solved. Problems can be simple (straightforward, easy to understand, with a clear solution) or complex (more challenging, with many parts or steps).

(ii) Algorithm

An algorithm is a set of clear, step-by-step instructions that help you solve a problem or complete a task. In other words, it is a set of rules to follow.

(iii) IF Statement

An IF statement is used to make decisions in a program. It tells the computer to do something only if a certain condition is true: the

computer checks whether a specific condition is true or false, and if it is true the computer does something; if it is false it does nothing. Example: 'If it is raining, then bring an umbrella.'

(iv) Identification of the problem

Identifying the problem is the process of figuring out what the problem is: understanding what is going wrong or what needs to be fixed. Example: you cannot find your favourite book.

(v) GOTO Statement

A GOTO statement is a shortcut that changes where the program is looking next: the computer moves to a different part of the program and starts running instructions from there. Example: in a game, choosing to skip to the next level moves you directly to that level.

Exercise 5. Differentiate the following (page 96)

(i) Problem and Algorithm

A problem is a situation or an issue that needs to be solved. An algorithm is the solution method: a set of clear, step-by-step instructions that helps you solve that problem or complete a task.

(ii) IF and GOTO statement

An IF statement makes a decision: it does something only if a certain condition is true, and does nothing if it is false. A GOTO statement does not test a condition by itself; it is a shortcut that jumps the program to a different part and continues running instructions from there.

(iii) Ordered and Unambiguous

Ordered means the steps are in the right sequence to get the correct result, for example you must mix ingredients before baking them. Unambiguous means there is no confusion about what each step means, for example instead of saying 'heat', specify 'heat on medium for 5 minutes'.

(iv) Program and Algorithm

An algorithm is a step-by-step procedure or set of rules to solve a specific problem or perform a task, written before coding. A program is the set of instructions written in a specific language that a computer can understand and execute; writing a program (coding) turns the algorithm into something the computer runs.

Exercise 6. Write short answers (page 96)

(i) What is problem solving?

Problem solving means figuring out what the problem is, gathering information, thinking of solutions (brainstorming), evaluating and choosing the best solution, implementing it, checking if it worked, and learning from the experience (reflecting).

(ii) Well Defined steps of algorithm

Well-defined steps means the algorithm has specific, detailed steps that are easy to follow. For example, instead of just saying 'make a sandwich', the steps should clearly say what to do, like 'spread peanut butter'.

(iii) What is the origin of the word Algorithm

The word 'algorithm' is derived from the name of the mathematician Mohammed ibn-Musa al-Khwarizmi (born 780 AD, Iraq; died 850 AD).

Q7. Write an Algorithm for making noodles for your friend. (Model answer)

Algorithm for making noodles for your friend. Step 1: Start. Step 2: Take a saucepan and pour fresh water into it. Step 3: Turn on the stove and boil the water. Step 4: Add the noodles to the boiling water. Step 5: Add the tastemaker (spices) and stir. Step 6: Cook until the noodles are soft. Step 7: Turn off the stove and serve the noodles to your friend. Step 8: End. (Model algorithm, following the book's tea/fries pattern; quantities and timing are your own choice.)

Q8. Write an Algorithm for Finding greater number between two values.

Algorithm for finding the greater number between two values. Step 1: Start. Step 2: Take two inputs (a and b) from the user. Step 3: If a is greater than b, then go to Step 4, otherwise go to Step 5. Step 4: Print 'a is greater than b', then go to Step 6. Step 5: Print 'b is greater than a'. Step 6: End. (This is the book's own example, section 5.4.)

Q9. Write an Algorithm for 2(3 + 5).

Algorithm for 2(3 + 5). Step 1: Start. Step 2: Follow the order of operations: brackets first, then multiplication. Step 3: Solve inside the brackets: 3 + 5 = 8. Step 4: Multiply: 2 x 8 = 16. Step 5: Print

the result, 16. Step 6: End. (Worked exactly like the book's example 2 x 2 + 2, where multiplication comes before addition.)

Q10. What are the advantages of Algorithm? How it is beneficial?

Algorithms are beneficial because they give a clear set of steps to follow, so a problem is broken into manageable steps and solved efficiently. Writing the algorithm before coding is a very beneficial habit: it resolves various errors and confusions, makes the steps easy to understand and test, and lets anyone follow the same steps to reach the same correct result.

Lab Activities (page 97)

Practical 1: Make an algorithm for making tea on a chart. Guidance: use the book's own 'Making Milk Tea' example (Start; place fresh water in a saucepan; turn on the stove; boil the water; put in the tea leaves; when the water colour changes to black add milk; add sugar; boil for some more time; serve; End) and present it neatly on a chart.

Practical 2: Make an algorithm for a table of 5 on a chart. Guidance: write it in the book's step pattern, for example: Step 1: Start. Step 2: Take the number 5 and a counter n = 1. Step 3: Print 5 x n. Step 4: Increase n by 1. Step 5: If n is less than or equal to 10, go back to Step 3, otherwise go to Step 6. Step 6: End. Present it on a chart.

Practical 3: Make an algorithm to go to school in a rainy day. Guidance: write it in steps with a condition, for example: Step 1: Start. Step 2: If it is raining, take an umbrella and raincoat, otherwise go without them. Step 3: Wear school shoes that will not slip. Step 4: Leave home early. Step 5: Go to school by the safe

route, avoiding flooded streets. Step 6: Reach school. Step 7: End. (Model algorithm; your own steps are acceptable.)

Unit 6: Programming

Exercise 2. Fill in the blanks (page 119) [the book numbers this exercise '2' before exercise '1']

i. A ____ is a set of well-defined instruction given to the computer to do a task.

program

ii. The most basic programming language, directly understood by the computer's hardware is called ____.

low level language (machine language)

iii. ____ uses more symbols and abbreviations in instructions.

Assembly language (middle level language)

iv. ____ are programming languages that are designed to be easy for humans to read and write.

High-level languages

v. JavaScript is used in ____.

websites and mobile application development

vi. Scratch is a tool developed by ____.

MIT Media Lab

vii. Scratch Programming deals with ____ with the help of blocks.

creating a script

viii. Multiple blocks become a program that will work as a ____.

single unit

ix. Scratch Interface consists of ____ sections.

3 (three)

x. A loop is a programming structure that ____ a set of instructions or code until a certain condition is met.

repeats

Exercise 1. Tick the correct choice (pages 119-120) [the book numbers this exercise '1' after exercise '2']

(i) A set of well-defined instruction given to the computer to do a task is called ____.

- Data
- Scratch
- ✓ Program
- MS Paint

(ii) A program which converts a high level language program into machine language is called ____.

- MS-Word
- ✓ Compiler
- High Level Language
- Middle level language

(iii) The most basic programming language, directly understood by the computer's hardware is referred to as ____.

- ✓ Low Level Languages
- Compiler
- High Level Language
- Middle level language

(iv) ____ a visual programming language and online community designed primarily for children and beginners to help them learn the basics of coding developed by MIT.

- MS-Word
- Paint 3D
- ✓ Scratch
- MS Paint

(v) In Scratch, the area where your project comes to life, like a theater stage where all the action takes place, is known as ____.

- ✓ Stage
- Scripts
- Blocks
- Sprites

(vi) A code element that you can simply drag into the stage to create a script is referred to as ____.

- Sprites
- Compiler
- ✓ Blocks
- Stage

(vii) ____ basic programming elements in Scratch.

- Sprites
- repeat
- ✓ Blocks
- Stage

(viii) 'Again and Again' or 'Repeat' is a concept that represents ____.

- ✓ Loop
- GOTO
- Compiler
- Stage

(ix) ____ is a common way to test and run Scratch projects.

- Repeat
- Stage
- ✓ Clicking green flag
- Sprites

(x) Scratch programming language was developed by ____.

- Adobe
- Microsoft
- Apple
- ✓ MIT

Note on MCQ (i): the book mislabels the options, printing (a) Data, (b) Scratch, then (a) Program, (b) MS Paint again. Read in order, the four options are Data, Scratch, Program, MS Paint, and the correct

one is Program.

Exercise 3. True or False - rewrite the statements to make them true (pages 120-121)

i. A code is a set of logical, mathematical and sequential instructions. True

ii. A Compiler converts a Middle language program into high level language. False

iii. language is called. (Flagged: the statement is incomplete as printed in the book; no true/false answer is possible.)

Flagged, not answered: item iii is printed incomplete in the book ('iii. language is called.') with no subject, so it cannot be answered true or false as printed.

iv. Scratch is a tool developed by Apple Media Lab (USA) in 2003. False

v. Scratch is used to make system software through animations and visual effects. False

vi. Sprite is at upper right (where actors act, and information is displayed). False

Corrected statements: (ii) A Compiler converts a HIGH LEVEL language program into MACHINE language. (iv) Scratch is a tool developed by the MIT Media Lab (USA) in 2003. (v) Scratch is used to make stories, animations and games through animations and

visual effects, not system software. (vi) The STAGE is where actors act and information is displayed; a sprite is an object/actor used for developing a program. Note on (i): the unit defines 'program' rather than 'code'; the statement is marked True as the standard meaning of code.

Exercise 4. What is a compiler? (page 121)

4. What is a complier? (compiler)

A compiler is a special program that translates code written in a high-level programming language into machine code, which is the set of instructions a computer's hardware can directly execute. It works in three steps: (1) Source Code: the programmer writes code in a high-level language like C++, Java or C#. (2) Compilation: the compiler processes this source code, checking for errors and converting it into machine code or an intermediate code. (3) Output: the result is usually an executable file (like .exe on Windows) that the computer can run. (The book spells this question 'complier'; the correct spelling is compiler.)

Exercise 5. Describe 5 high level languages and their usage (page 121)

5. Describe 5 high level languages and their usage.

(1) Python: used in Artificial Intelligence, data science, web development, automation and scripting, and social media applications; for example Instagram uses Django (a Python framework) for its web applications. (2) JavaScript: used in websites and mobile application development; for example Gmail and Google Maps use JavaScript for smooth, seamless user experiences. (3) Java: used for internet programming and Android mobile

applications. (4) C#: used for mobile devices and video games; for example 'Angry Birds 2' and 'Monument Valley' use C# for game logic and interactions. (5) C/C++: C is used for hardware programming such as automobiles and medical devices, for example microcontrollers and embedded systems.

Exercise 6. Define Low level language (page 121)

6. Define Low level language.

Low level language (machine language) is the most basic programming language, directly understood by the computer's hardware. It consists of binary code (0s and 1s), for example 10101000, and it is very difficult to learn.

Exercise 7. Explain the following terms (page 121)

(a) Sprites

Sprites are objects/actors used for developing a program in Scratch. Each object in Scratch is called a sprite; you can choose one from the library, paint your own, upload an image, or take a picture from a webcam.

(b) Scripts

A script is the name of a program or method in Scratch; it tells the actor (sprite) what to do. Scratch programming deals with creating a script with the help of blocks.

(c) Block

Blocks are the units in Scratch programming. Each block has its own set of functions and properties and is unique from the others;

multiple blocks become a program that will work as a single unit.

Exercise 8. Name the blocks for the following tasks (page 121)

i. To print a message in a bubble:

Say block (from the Looks category): 'say ... for ... secs', for example 'say Hello! for 2 secs'.

ii. To perform a task when green flag is clicked:

'When green flag clicked' block (from the Events category), snapped on top of the stack; whenever you click the green flag, your script will start. To stop, click the red stop button.

iii. To draw a line behind the sprite:

Pen down block (from the Pen category): with the pen down, the sprite draws a line behind itself as it moves.

iv. To say Hello for few seconds:

Say block (from the Looks category): 'say Hello for 2 secs' (type Hello and the number of seconds in the block).

v. To move the sprite few steps:

Move block (from the Motion category): 'move 10 steps' (change the number for more or fewer steps).

vi. To play drums:

Play drum block (from the Sound category): 'play drum 1 for 0.25 beats' (choose a drum from the pull-down menu: snare drum, bass drum, side stick, crash cymbal, open hi-hat).

Exercise 8 (again). Write a script to do the following (page 121)

Write a script to: (i) Double the size of the sprite, (ii) Wait 2 seconds, (iii) Change the color of the sprite to green, (iv) Wait 4 seconds, (v) Change the whirl effect to 100, (vi) Say 'that's all folks'.

Script (top to bottom, snapped into one stack; run it with the green flag): (1) change size by 100 (this doubles the sprite, from 100% to 200%). (2) wait 2 seconds (from the Control category). (3) set color effect to green (from the Looks category; the book's colour blocks are 'set color effect' and 'change color effect by'). (4) wait 4 seconds. (5) set whirl effect to 100 (from the Looks category). (6) say 'that's all folks' (from the Looks category).

Lab Activities (page 121)

Practical 1: Make a chart of types of languages, and describe the high level languages and their usage. Guidance: divide the chart into low level (machine language, binary 0s and 1s), middle level (assembly language, symbols and abbreviations) and high level languages (Python, JavaScript, Java, C#, C/C++, Go, R, Swift, PHP, Scratch) with one use for each.

Practical 2: Find the different building blocks to make Scratch code and discover the logic behind interlocking. Guidance: open Scratch, drag blocks from the Blocks Palette into the Scripts Area, and notice how blocks snap together (interlock) into a stack that runs as a single unit.

Practical 3: Program a sprite: drag blocks from the Blocks Palette to the Scripts Area. Guidance: for example drag 'move 10 steps' and

'play drum 1 for 0.25 beats' into the Scripts Area, snap them together, and click the stack (or the green flag) to run it.

Practical 4: Discover some blocks which have pull-down menus. Guidance: for example the 'play drum' block has a pull-down menu of drums (snare drum, bass drum, side stick, crash cymbal, open hi-hat), and the 'when key pressed' block has a pull-down menu of keys (space, up/down/left/right arrows).

Practical 5: Focus on helping students to develop storytelling skills through a variety of activities, providing opportunities to work together and build on each other's work. Guidance: work in a group to plan a short Scratch story, divide the scenes among yourselves, build each scene with sprites and scripts, and combine them into one project.

Unit 7: Digital Citizenship

Exercise 1. Fill in the blanks (page 130)

(i) ____ refers to the responsible and ethical use of technology by individuals when interacting online.

Digital Citizenship

(ii) ____ refers to the principles and moral guidelines that govern the responsible use of technology and the internet.

Digital Ethics

(iii) ____ is a law that helps protect creative works and ideas.

Copyright

(iv) ____ is the act of using someone else's work, ideas, or expressions without proper acknowledgment.

Plagiarism

(v) ____ refers to the unauthorized use, distribution, or reproduction of copyrighted materials.

Piracy

(vi) ____ causes painful swelling of joints such as shoulder, wrist and fingers.

RSI (Repetitive Strain Injury)

(vii) Both back and neck ache can be caused by ____.

sitting incorrectly (poor posture)

(viii) ____ starts when staring at a computer screen for too long.

Eye strain

(ix) If you press the ____ directly to close your computer, it will likely only put your machine in sleep or hibernation mode.

power button

(x) Before using a USB drive or external hard drive on your computer, always scan it with ____.

antivirus software

Exercise 2. Tick the correct choice (pages 131-132)

(i) ____ is the act of using someone else's work, ideas, or expressions without proper acknowledgment, presenting them as your own.

- Internet
- ✓ Plagiarism
- Phishing
- Digital citizenship

(ii) Before using a USB drive or external hard drive on your computer, always scan it with ____.

- Copyright
- MS Paint
- ✓ antivirus software

MS PowerPoint

(iii) If you have kept several tabs open in your browser that you don't want to lose, simply use ____ so you can come back to them later.

✓ bookmarks

MS PowerPoint

Piracy

Antivirus

(iv) ____ is a legal protection granted to the creators of original works, such as books, music, and software, giving them exclusive rights to use and distribute their creations.

Digital citizenship

Cyberbullying

Browser

✓ Copyright

(v) ____ must be installed in the computer to save from virus.

✓ Antivirus

Program

Windows

All of these

(vi) ____ authentication (2FA) to add an extra layer of security to your email account.

Antivirus

✓ Use two-factor

password

Google Chrom

(vii) ____ causes painful swelling of joints such as shoulder, wrist and fingers.

Confidentiality

Security

✓ RSI

Copyright

(viii) ____ starts when staring at a computer screen for too long.

✓ Eye strain

MS Paint

Social Media

Scratch

(ix) Both back and neck ache can be caused by ____ or using a chair without back rests.

using Internet

using social media

✓ sitting incorrectly

phone

(x) You should not open emails and ____ from suspicious source.

MS Word

✓ attachments

scratch

windows

Exercise 3. Define the following terms (page 132)

(a) Copyright

Copyright is a law that helps protect creative works and ideas. It means that if you create something original, like a drawing, a story, or a song, you have the right to decide how other people can use it, and others cannot use it without asking you first. It is a legal protection granted to creators, giving them exclusive rights to use and distribute their creations.

(b) Plagiarism

Plagiarism is the act of using someone else's work, ideas, or expressions without proper acknowledgment, presenting them as your own. Examples: copying text from a book, article or website without giving credit; paraphrasing someone else's ideas without citing the source; submitting someone else's work as your own. It can lead to academic penalties, loss of credibility, legal consequences and damage to one's reputation. Avoid it by giving credit, using your own words, and citing sources.

(c) Piracy

Piracy refers to the unauthorized use, distribution, or reproduction of copyrighted materials. It involves obtaining or sharing content such as music, movies, software or books without permission from the copyright holder. Piracy is illegal and unethical because it deprives creators and copyright owners of revenue and recognition.

(d) Digital citizenship

Digital Citizenship refers to the responsible and ethical use of technology by individuals when interacting online. Its key aspects are: online safety (protecting personal information, understanding privacy settings, being aware of scams, phishing and cyberbullying); respectful communication; ethical use of information (citing sources, avoiding plagiarism, respecting copyright); digital literacy; responsible participation; and balancing technology use (managing screen time and health effects).

(e) RSI

RSI stands for Repetitive Strain Injury. It causes painful swelling of joints such as shoulder, wrist and fingers, for example from repeatedly clicking a mouse button or moving fingers on the keyboard. Prevention: take breaks to rest your hands, use an ergonomic keyboard and mouse, use a wrist rest pad, and use enough space to work comfortably.

Exercise 4. Identify the negative impact of ICT devices on health (page 132)

4. Identify the negative impact of ICT devices on health.

(1) Repetitive Strain Injury (RSI): painful swelling of joints such as shoulder, wrist and fingers, from repeated mouse clicking or keyboard use. Prevention: rest your hands with breaks, use an ergonomic keyboard and mouse, use a wrist rest pad, and keep enough space to work comfortably. (2) Back and neck problems: ache caused by sitting incorrectly (poor posture) or using a chair without back rests. Prevention: take regular breaks to stretch, use adjustable chairs, sit upright against the back rest, and keep your

feet flat on the floor or use a footrest. (3) Eye strain and headaches: eye strain starts when staring at a computer screen for too long and can blur your vision. Prevention: take regular breaks every hour to rest your eyes, use TFT screens rather than CRT, use an anti-glare screen, ensure good room light, and have your eyes tested regularly.

Unit 8: Entrepreneurship in Digital Age

Exercise 1. Fill in the blanks (pages 141-142)

(i) ____ is the process of starting and running a new business or venture.

Entrepreneurship

(ii) The ____ relies on physical locations, face-to-face interactions.

Traditional Entrepreneurship

(iii) The ____ utilizes raw materials to create products that are then sold in the market.

Manufacturer

(iv) Traditional business model needs ____ and other utilities.

place, furniture, transport, staff

(v) Digital businesses work ____; the traditional business model has restrictions on when and where they function.

24/7

(vi) Digital entrepreneurship refers to the process of creating and managing a business ____.

primarily online

(vii) The first step of entrepreneurship process is ____.

Idea Generation

(viii) E-commerce refers to the buying and selling of goods and services over the ____.
internet

(ix) Amazon started as a small ____ in 1994.
online bookstore

(x) ____ founded by Richard Liu (Liu Qiangdong) in 1998.
Jingdong (JD.com)

(xi) The owner and founder of ____ is Jack Ma, a Chinese entrepreneur.
Alibaba (Alibaba Group)

Note: the book misnumbers the fill items, printing (i) three times and (vii) twice; the items above are given in the book's order with corrected sequential numbering.

Exercise 2. Tick the correct choice (page 142)

(i) ____ is the smart application of the power of digital technologies to business.

- ✓ Digital entrepreneurship
- vision
- Technology
- strategy

(ii) Which of the following is not component on digital

entrepreneurship?

✓ Digital entrepreneurship

vision

technology

strategy

(iii) ____ is the creator of Facebook.

✓ Mark Zuckerberg

Bill Gates

Steve Jobs

Travis Kalanick

(iv) ____ introduced Uber.

Mark Zuckerberg

Bill Gates

Steve Jobs

✓ Travis Kalanick

(v) ____ uses social media, phones, and other electronic devices to advertise a product or service.

✓ Digital Promotion

proposition

Quality Assurance

All of these

Note on MCQ (v): the unit's own term is Digital Marketing, defined as the strategies used to promote products and attract customers online. Among the printed options, (a) Digital Promotion is the

intended answer.

Exercise 3. Answer the following questions (page 142)

(a) Define Digital Entrepreneurship.

Digital entrepreneurship refers to the process of creating and managing a business primarily online, using digital technologies and the internet to deliver products or services. Digital entrepreneurs leverage digital platforms such as websites, social media, e-commerce and mobile apps to reach customers, market their offerings and operate their businesses. In short, it is the smart application of the power of digital technologies to business. Its three principal components are the vision, the technology and the strategy.

(b) Define Traditional Entrepreneurship.

Traditional entrepreneurship relies on physical locations, face-to-face interactions and conventional business practices. Examples include retail stores, shops, restaurants and manufacturing companies. Customers interact with the business in person or via phone, with a focus on local or regional markets, and it primarily uses traditional marketing channels such as print ads, television, radio and in-person sales.

Exercise 4. Differentiate Traditional and Digital Entrepreneurship (page 142)

4. Differentiate Traditional and Digital Entrepreneurship.

Traditional business needs more capital, and needs place, furniture, transport, staff and other utilities; entering the market, production and storage, and distribution are difficult, and work is slower.

Digital business is cost-effective in this respect, entering the market and distribution are easy, work is faster, and the workplace is digital rather than physical. The digital approach is convenient for customers (for example Amazon's extensive product catalog), while rigidity is a major issue in the traditional setup. Digital businesses work 24/7, while the traditional model has fixed timings and less flexible customer service. Communication in traditional business is face to face, while digital business communicates through computers and digital technologies. Traditional organizational structure is hierarchical; digital structure is flexible and collaborative. One limit of digital business: you cannot try a product before buying it (for example a mobile phone from Amazon); you rely on customer reviews and specifications, while the traditional model overcomes this obstacle.

Exercise 5. Describe the Areas of Digital Entrepreneurship (page 142)

5. Describe the Areas of Digital Entrepreneurship.

Digital entrepreneurship can be applied to various sectors: (1) Agriculture: using digital platforms to connect farmers with buyers, smart farming through sensors and data analytics, or drones for crop monitoring and precision agriculture. (2) Trading: online trading platforms for buying and selling stocks or commodities, often using algorithms and data analysis. (3) Manufacturing: digital tools for automation, like robotics and Internet of Things (IoT) devices, to improve efficiency in production lines, and digital twins to simulate and optimize processes. (4) Education: online learning platforms such as e-learning courses or educational apps, with

virtual classrooms and digital resources. (5) Real Estate: digital platforms to list properties, conduct virtual tours, and connect buyers with agents, with data analytics for market analysis and pricing.

Exercise 6. What is E-Commerce (page 142)

6. What is E-Commerce.

E-commerce (electronic commerce) refers to the buying and selling of goods and services over the internet, through digital platforms rather than traditional brick-and-mortar stores. Its key aspects are: online stores (websites or apps where businesses and individuals list and sell products, for example Amazon, eBay, Alibaba and Shopify stores); payment systems (credit/debit cards, digital wallets like PayPal and Alipay, and bank transfers, with secure payment gateways); shopping experience (product searches, reviews, ratings and personalized recommendations; customers can view products, compare prices and order from anywhere in the world); order fulfillment (storing, packaging and shipping products, through the business's own logistics or third-party fulfillment centers); customer service (live chat, email support and help centers); and digital marketing (strategies to promote products and attract customers online).

Exercise 7. What is Entrepreneurship Process (page 142)

7. What is Entrepreneurship Process.

The entrepreneurship process (how to start a business) has six steps: (1) Idea Generation: come up with a new idea for a product or service, something that solves a problem, makes life easier, or is

just fun. (2) Research: learn more about the idea; ask 'Will people like this?' or 'Is anyone else doing this?' to understand if the idea is good and how to make it better. (3) Planning: make a plan for how the business will work, what you need, how much money it will cost, and how you will tell people about the idea; the plan is like a roadmap. (4) Start the Business: put the plan into action, create the product or service and start selling it. (5) Manage and Grow: keep customers happy, keep track of money, and think of ways to make the business even better. (6) Evaluate and Improve: look at how the business is going; if something is not working, think of ways to fix it, and always look for ways to improve and grow.

Exercise 8. How amazon.com started? (page 142)

8. How amazon.com started?

Amazon started as a small online bookstore in 1994. It was created by Jeff Bezos, who wanted to sell books to people all over the world using the internet; he began the business in his garage in Seattle, Washington. At first Amazon only sold books, but it quickly became popular because people liked how easy it was to buy books online. As the company grew, Amazon added music, movies, electronics and toys, and it is now known for fast shipping, a wide selection, and being a platform where millions of people shop every day.